
Exercise sheet N°2: Relations and functions

Exercise 1.1

Let $\mathcal{R}$ be a relation defined on $\mathbb{R}$ as follows:

$$x\mathcal{R}y \Leftrightarrow x^3 - y^3 = 3(x - y)$$

1. Show that $\mathcal{R}$ is an equivalence relation.
2. Determine $cl(0)$, $cl(-2)$ and $cl(3)$.
3. Let $a \in \mathbb{R}$. Determine $|cl(a)|$ according to the values of a . $|cl(a)| = Card(cl(a))$.

Exercise 1.2

Let $\mathcal{R}$ be a relation defined on $\mathbb{Z}$ as follows:

$$x\mathcal{R}y \Leftrightarrow x^2 - y^2 = 7x - 7y$$

1. Show that $\mathcal{R}$ is an equivalence relation.
2. Determine $cl(0)$, $cl(1)$ and $cl(3)$.
3. Let $a \in \mathbb{Z}$. Determine $|cl(a)|$ according to the values of a .

Exercise 2.1

We define the relation $\mathfrak{R}$ on $\mathbb{Z}^* \times \mathbb{Z}^*$ as follows:

$$(a, b)\mathfrak{R}(c, d) \Leftrightarrow ad = cb$$

1. Prove that $\mathfrak{R}$ is an equivalence relation.
2. Determine $cl(2, 4)$.

Exercise 2.2

We define the relation $\mathfrak{R}$ on $\mathbb{Z}^2$ as follows:

$$(a, b)\mathfrak{R}(c, d) \Leftrightarrow ab = cd$$

1. Prove that $\mathfrak{R}$ is an equivalence relation.
2. Determine $cl(1, 1)$ and $cl(0, 0)$.

Exercise 3.1

We define on $\mathbb{N}^*$ the relation $\mathcal{R}$ as follows:

$$x\mathcal{R}y \Leftrightarrow \exists n \in \mathbb{N}^*, y = x^n$$

1. Show that $\mathcal{R}$ is an order relation.
2. Is this order total?

Exercise 3.2

We define on $\mathbb{N}^*$ the relation $\mathcal{R}$ as follows:

$$x\mathcal{R}y \Leftrightarrow \exists n \in \mathbb{N}^*, y = nx$$

1. Show that $\mathcal{R}$ is an order relation.
 2. Is this order total?
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Exercise 4.1

We define the relation $\mathcal{R}$ on $\mathbb{R}^2$ as follows:

$$(a, b)\mathcal{R}(c, d) \Leftrightarrow |a - c| \leq d - b$$

1. Show that $\mathcal{R}$ is an order relation
2. Is this order total?

Exercise 4.2

We define the relation $\mathcal{R}$ on $\mathbb{R}^2$ as follows:

$$(a, b)\mathcal{R}(c, d) \Leftrightarrow a \leq c \text{ and } b \leq d$$

1. Show that $\mathcal{R}$ is an order relation
2. Is this order total?

Exercise 5 (*)

On $\mathbb{R}_+^*$, we define the relation $\mathcal{R}$ by:

$$x\mathcal{R}y \Leftrightarrow e^{\frac{1}{x}} \leq e^{\frac{1}{y}}$$

Prove that $\mathcal{R}$ is a total order relation.

Exercise 6

Consider the function f from $[0,1]$ to $[0,2]$ defined by: $f(x) = 2 - x$,

and the function g from $[-1,1]$ to $[0,2]$ defined by: $g(x) = x^2 + 1$.

1. Determine:

$$f\left(\frac{1}{2}\right), f\left(\left\{0, \frac{2}{3}, 1\right\}\right), f^{-1}(0), f^{-1}\left(\left\{1, \frac{5}{3}\right\}\right), g(\{-1, 1\}), g([-1, 1]), \text{ and } g^{-1}([0, 2]).$$

2. Is the function f bijective? Justify.
3. Is the function g bijective? Justify.

Exercise 7

I. Let f be a function defined as follows:

$$\begin{aligned} f : \mathbb{R}^2 &\rightarrow \mathbb{R}^2 \\ (x, y) &\mapsto f(x, y) = (x + y, xy) \end{aligned}$$

1. Calculate $f(2, 3)$ and $f(3, 2)$.
2. Is there an element $(a, b) \in \mathbb{R}^2$ such that $f(a, b) = (3, 5)$?
3. Is the function f injective? Surjective? Bijective? Justify

II. Let g be a function defined as follows:

$$\begin{aligned} g : \mathbb{R}^2 &\rightarrow \mathbb{R}^2 \\ (x, y) &\mapsto g(x, y) = (x + y, x - y) \end{aligned}$$

Show that the function g is bijective.

Exercise 8.1

We define the function f as follows:

$$f: \mathbb{R} \rightarrow]1, +\infty[$$

$$x \mapsto f(x) = \sqrt{1 + e^x}.$$

Show that f is bijective and determine its inverse function f^{-1} .

Exercise 8.2

We define the function f as follows:

$$f: \mathbb{R} \rightarrow]1, +\infty[$$

$$x \mapsto f(x) = \frac{1}{2}e^x + 1.$$

Show that f is bijective and determine its inverse function f^{-1} .

Exercise 9 (*)

1. Let f be a function defined from $\mathbb{R} \setminus \{-1, 1\}$ to $\mathbb{R}$ by

$$f(x) = \frac{x^2}{x^2 - 1}$$

Is f bijective? Justify

2. We define the two functions g and h as follows:

$$g: \mathbb{R}_+ - \{1\} \rightarrow]-\infty, 0] \cup]1, +\infty[\quad h:]-\infty, 0] \cup]1, +\infty[\rightarrow \mathbb{R}_+ - \{1\}$$

$$x \mapsto g(x) = f(x) \quad x \mapsto h(x) = \sqrt{\frac{x}{x-1}}$$

a. Determine $g \circ h$ and $h \circ g$.

b. Conclude that g is bijective and provide its inverse f^{-1} .

(*) Additional (brief solution of unrepeatable ideas)